

Organizational Depth at Finite Time: A Fixed-Resolution No-Go Boundary

Abstract

When does infinite organizational depth represent a physically meaningful finite-time escape rather than increasingly fine indexing of a finite transformation? We reformulate this question within Organizational Accessibility around three results. First, a sequential stochastic accessibility process with transition rate $\lambda_n = N_n \nu_n p_n$ is non-explosive when total material is bounded, each independently counted realization requires a positive minimum material allocation, and candidate generation per realization is uniformly bounded. Second, a continuum cascade can retain bounded aggregate throughput while its specific turnover diverges, because the inventory assigned to successive levels tends to zero. This calculation identifies a shrinking-scale possibility but does not derive its accelerating kinetics physically.

The central result is a conditional fixed-resolution no-go theorem. If transition costs and durations satisfy $\varepsilon_n \tau_n \geq c_n d_{\text{phys},n}^2$, finite total charged energy and time imply $\sum_n \sqrt{c_n} d_{\text{phys},n} < \infty$. A uniform positive kinetic coefficient therefore bounds accumulated physical distance. If physical distance also uniformly controls operational distinguishability, only finitely many transitions at any fixed operational resolution can occur within finite time and energy. A constant-speed overdamped example saturates the bound while crossing infinitely many refinement labels before reaching an ordinary finite state.

These are elementary mathematical results with conditional physical applications, not a universal prohibition of singular dynamics. The remaining question is whether an autonomous system can preserve operationally nontrivial retained transformations while its physical speed–distance cost scale collapses, without concealing an unbounded kinetic, control, memory, or measurement resource.

1 Introduction: from singularity to physical distinguishability

Organizational Accessibility asks how currently realized organization changes which successor organizations can be generated, established, and retained [1]. Its central historical claim is conditional: previously successful organization can become infrastructure for subsequent organization. This does not imply that organizational change accelerates indefinitely, or that increasing organizational depth corresponds to increasing physical change.

A proposed finite-time hierarchy introduces a further question. Suppose retained states are labelled

$$x_0, x_1, x_2, \dots$$

and their transition times accumulate before a finite deadline. What makes these labels represent distinct physical organizations? Without an answer, an infinite hierarchy may describe a finite trajectory divided into increasingly small pieces.

The present paper therefore asks:

Under what conditions can infinite organizational depth remain physically meaningful at finite time, finite charged energy, and fixed operational resolution?

The analysis separates three issues. A stochastic event model requires a sufficient rate of completed transitions. A continuum throughput model requires an interpretation of increasingly

rapid turnover of shrinking quantities. An organizational interpretation additionally requires physically realizable distinctions between successive retained states.

The resulting boundary is conditional. Finite time and energy exclude infinitely many fixed-resolution transitions when an appropriate speed–distance inequality holds uniformly and its physical metric controls operational distinguishability. Neither condition follows merely from assigning an increasing organizational level number.

The generic mathematical ingredients are established: pure-birth non-explosion, energy accounting, Cauchy–Schwarz inequalities, thermodynamic speed limits, and metric convergence [2, 7, 8, 12]. The proposed contribution is their application to the interpretation of organizational depth, including an explicit separation of kinetic degeneracy from degenerating physical resolution.

2 Organization as amount, configuration, and operational distinction

Three descriptions of organization must be distinguished.

Organization as amount is an extensive quantity whose increase requires an increase in a material or energetic stock. The component abundances in the resource-limited model of Organizational Accessibility have this interpretation.

Organization as configuration or history concerns arrangement, architecture, or retained causal structure. Such quantities need not increase in direct proportion to material stock. The same physical substrate may participate in many successive configurations.

Organization as an operational distinction concerns differences between retained states that can be resolved under a specified measurement standard. Different labels do not establish such a distinction.

The physical state description must include the subsystems needed to implement the claimed transformation. These can include processors, catalysts, controllers, memory, interfaces, fuel stores, and retained records. Omitting them can turn a physical cost into an apparently cost-free external condition.

Transitions and accounting

Set $t_0 = 0$ and write

$$t_{n+1} = t_n + \tau_n, \quad \tau_n > 0.$$

A finite-time sequential accumulation has

$$T_* = \sum_{n=0}^{\infty} \tau_n < \infty. \quad (2.1)$$

Let $\varepsilon_n \geq 0$ be a *charged energetic cost*. This means a nonnegative expenditure that can legitimately be counted against a cumulative budget:

$$\sum_{n: t_{n+1} \leq T} \varepsilon_n \leq B(T) < \infty \quad \text{for every finite } T. \quad (2.2)$$

Raw signed work, heat, stored energy, and excess dissipation are not interchangeable choices for ε_n . The selected cost must match both the budget and the speed–distance inequality used below.

For example, a model may justify

$$B(T) = E_0 + \int_0^T P_{av}(t) dt$$

after accounting for finite initial stores and the usable free-energy sources that pay the charges. Bounded incoming power alone does not bound gross internal processing power, nor does it justify charging the same recoverable energy repeatedly as if it were irreversibly consumed.

The finite-action results below are statements about sequences satisfying explicit inequalities. They apply pathwise when their premises hold pathwise. They can also apply to prescribed sequences of thermodynamic protocols with deterministic durations and ensemble-mean costs, provided the same ensemble interpretation is used throughout. They do not convert an average thermodynamic inequality into an almost-sure event-cost bound.

Material concentration is a separate question

The normalized finite-dimensional production model in Organizational Accessibility has a material balance of the form $\dot{M} = J - M$. Its abundance bounds concern organization as amount. Spatial extensions raise a different regularity question: control of integrated material does not automatically control pointwise concentration or stronger norms [14].

The present fixed-resolution results neither establish nor exclude all such spatial singularities. They concern specified sequences of retained transformations, their physical costs, and their operational interpretation.

3 Closure of stochastic route C

Consider a sequential pure-birth accessibility process

$$O_0 \longrightarrow O_1 \longrightarrow O_2 \longrightarrow \cdots.$$

At level n , let

$$\lambda_n = N_n \nu_n p_n, \tag{3.1}$$

where N_n is the number of active searching realizations, ν_n is their candidate-generation rate per realization, and p_n is the probability that a candidate belongs to the specified successor class and establishes.

Because p_n is a probability,

$$0 \leq p_n \leq 1.$$

This bound is part of the model definition, not an additional empirical hypothesis.

Theorem 1 (Closure of stochastic accessibility). *Suppose, at every reachable level,*

$$M_n \leq M_* < \infty, \quad M_n \geq m_{\min} N_n, \quad m_{\min} > 0,$$

and

$$0 \leq \nu_n \leq \nu_{\max} < \infty.$$

Then the pure-birth process with rates (3.1) is non-explosive.

Proof. The material inequalities give

$$N_n \leq \frac{M_n}{m_{\min}} \leq \frac{M_*}{m_{\min}}.$$

Therefore

$$\lambda_n \leq \Lambda := \frac{M_*}{m_{\min}} \nu_{\max} < \infty. \tag{3.2}$$

If $\Lambda = 0$, no transition occurs. If a reachable $\lambda_n = 0$, the sequential process stops there.

Otherwise,

$$\sum_{n=0}^{\infty} \frac{1}{\lambda_n} \geq \sum_{n=0}^{\infty} \frac{1}{\Lambda} = \infty.$$

This meets the classical non-explosion criterion for the pure-birth model [2].

A direct almost-sure argument is also available. Represent the independent holding times as

$$X_n = \frac{E_n}{\lambda_n}, \quad E_n \stackrel{\text{iid}}{\sim} \text{Exp}(1).$$

By (3.2),

$$X_n \geq \frac{E_n}{\Lambda}.$$

The strong law gives $\sum_n E_n = \infty$ almost surely, and hence

$$\sum_n X_n = \infty \quad \text{almost surely.}$$

□

The additive processor inequality $M_n \geq m_{\min} N_n$ is substantive. It is not valid if several overlapping descriptions of the same physical machinery are counted as independent full-rate realizations.

For the reciprocal-rate series to converge, it is necessary that $\lambda_n \rightarrow \infty$. Since $p_n \leq 1$, this requires $N_n \nu_n \rightarrow \infty$. Thus a C-type explosion must abandon at least one uniform bound or accounting assumption used in Theorem 1: bounded material, a positive minimum allocation per realization, bounded generation rate per realization, or additive processor accounting. In parameterized limits these failures may appear as

$$M_* \rightarrow \infty, \quad m_{\min} \downarrow 0, \quad \nu_{\max} \rightarrow \infty.$$

Alternatively, a model may depart from the sequential one-step event description, for example by allowing an unbounded depth increment within one event.

Failure of an assumption is not sufficient for explosion. Unbounded rates can still have a divergent reciprocal series.

Endogenizing N_n, ν_n, p_n remains relevant to finite accessibility rates and probabilities. It is not an unresolved route to explosion while the bounds of Theorem 1 are retained.

4 Why B survives bounded throughput: specific turnover and shrinking inventory

Consider the continuum cascade

$$\Phi_n = \kappa_n e_n^m, \quad m > 1, \tag{4.1}$$

$$\dot{e}_0 = J - \Phi_0, \tag{4.2}$$

$$\dot{e}_n = \eta \Phi_{n-1} - \Phi_n, \quad n \geq 1, \tag{4.3}$$

where $J > 0$, $0 < \eta \leq 1$, and e_n is the resource inventory assigned to level n .

For $e_n > 0$, define specific turnover and inventory residence time:

$$\rho_n = \frac{\Phi_n}{e_n} = \kappa_n e_n^{m-1}, \quad \tau_n^{\text{res}} = \frac{1}{\rho_n}. \tag{4.4}$$

The residence-time symbol is distinguished from the actual retained-transition duration τ_n . Identifying the two requires a model of what constitutes completion of the transformation.

Theorem 2 (Bounded throughput and unbounded specific turnover). *The following statements hold.*

1. If $0 < \rho_n \leq \rho_{\max} < \infty$, then

$$\sum_n \tau_n^{\text{res}} = \infty.$$

The same conclusion holds for actual durations satisfying $\tau_n \geq a/\rho_n$ with a constant $a > 0$.

2. For $\kappa_n = \kappa_0 b^n$, with $\kappa_0, b > 0$, the forced stationary profile is

$$\Phi_n^* = J\eta^n, \quad e_n^* = \left(\frac{J}{\kappa_0}\right)^{1/m} \left(\frac{\eta}{b}\right)^{n/m}.$$

Its specific turnover is

$$\rho_n^* = \kappa_0^{1/m} J^{(m-1)/m} (b\eta^{m-1})^{n/m}. \quad (4.5)$$

The stationary residence-time series converges exactly when

$$b\eta^{m-1} > 1.$$

3. If $0 < \eta < 1$ and $b\eta^{m-1} > 1$, both $\sum_n e_n^*$ and $\sum_n \Phi_n^*$ are finite, although $\rho_n^* \rightarrow \infty$.

Proof. For the first statement, $\tau_n^{\text{res}} \geq 1/\rho_{\max}$.

At stationarity, $\Phi_0^* = J$ and $\Phi_n^* = \eta\Phi_{n-1}^*$, giving $\Phi_n^* = J\eta^n$. Substituting this into (4.1) gives the inventory formula. Dividing throughput by inventory gives (4.5). The residence times form a geometric series with ratio $(b\eta^{m-1})^{-1/m}$.

Finally, for $0 < \eta < 1$,

$$\sum_n \Phi_n^* = \frac{J}{1-\eta} < \infty.$$

The supercritical specific-turnover condition implies $b > \eta^{1-m} > 1 > \eta$, so $(\eta/b)^{1/m} < 1$ and the inventory series also converges. $\square$

Thus

bounded aggregate throughput does not imply bounded specific turnover.

(4.6)

The distinction is a shrinking denominator. The rate Φ_n/e_n can increase while the physical amount processed at each level decreases.

For example, $m = 2$, $\eta = 1/2$, and $b = 4$ give finite total stationary inventory and aggregate throughput, while ρ_n^* grows proportionally to $(\sqrt{2})^n$. If e_n is an energy inventory and $1 - \eta$ is the dissipated fraction per handoff, stationary dissipation is

$$(1 - \eta) \sum_n \Phi_n^* = J.$$

This accounts for the resource transfers in the cascade. It does not account automatically for the apparatus needed to realize its increasingly large kinetic coefficients.

A power bound likewise need not control the ratio. If processing requires work $w_0 > 0$ per unit throughput, then

$$w_0 \Phi_n \leq P_{\max} \implies \rho_n \leq \frac{P_{\max}}{w_0 e_n},$$

which is not uniform when $e_n \rightarrow 0$.

Conversely, a model that establishes $\Phi_n \leq k_{\max} C_n$ and $C_n/e_n \leq \chi_{\max}$ obtains $\rho_n \leq k_{\max} \chi_{\max}$. Here C_n must measure physical processing capacity in compatible units. A global bound on C_n alone does not bound its ratio to a vanishing inventory.

Theorem 2 is a stationary model calculation. It neither proves that a transient solution reaches infinite retained depth in finite time nor derives κ_n from retained architecture. An infinite stationary profile already contains nonzero inventory at every level.

The physical question is therefore:

What permits successive retained transformations to occur at increasing specific rate while the amount being transformed shrinks?

The simple processor-count argument that closes C does not answer this continuum question. Neither does prescribing κ_n .

5 The finite-action theorem

Let the physical states belong to a metric space (X, d_{phys}) , and write

$$d_{\text{phys},n} = d_{\text{phys}}(x_n, x_{n+1}).$$

Suppose each transition obeys a speed–distance inequality

$$\varepsilon_n \tau_n \geq c_n d_{\text{phys},n}^2. \quad (5.1)$$

The physical meaning of this inequality must be justified for the chosen class of dynamics. Its costs, metric, and coefficients must be specified together. The theorem below is an exact mathematical consequence of the inequality; its use as a physical no-go theorem is conditional on that justification.

Theorem 3 (Finite-action bound). *Suppose (5.1) holds with $c_n \geq c_* > 0$, and*

$$\mathcal{E} := \sum_n \varepsilon_n < \infty, \quad \mathcal{T} := \sum_n \tau_n < \infty.$$

Then

$$\boxed{\sum_n d_{\text{phys},n} \leq \frac{1}{\sqrt{c_*}} \sqrt{\mathcal{E}\mathcal{T}} < \infty.} \quad (5.2)$$

Proof. For every finite k ,

$$\begin{aligned} \sum_{n < k} d_{\text{phys},n} &\leq \frac{1}{\sqrt{c_*}} \sum_{n < k} \sqrt{\varepsilon_n \tau_n} \\ &\leq \frac{1}{\sqrt{c_*}} \sqrt{\left(\sum_{n < k} \varepsilon_n \right) \left(\sum_{n < k} \tau_n \right)} \\ &\leq \frac{1}{\sqrt{c_*}} \sqrt{\mathcal{E}\mathcal{T}}. \end{aligned}$$

The second inequality is Cauchy–Schwarz. Taking the limit of the nondecreasing partial sums proves the result. $\square$

Consequently,

finite charged energy + finite time + uniform speed–distance control

implies finite accumulated physical distance.

No assumption of monotonic organizational improvement is used. The theorem applies equally to productive changes, reversals, oscillations, or unsuccessful transformations, provided the stated physical inequalities apply to the counted transitions.

6 The fixed-resolution no-go corollary

For $\delta > 0$, let

$$K_\delta^{\text{phys}}(T) = \# \{n : t_{n+1} \leq T, d_{\text{phys},n} \geq \delta\}.$$

Corollary 1 (Fixed physical resolution). *Under the uniform-coefficient assumptions of Theorem 3 and the budget (2.2),*

$$K_\delta^{\text{phys}}(T) \leq \frac{1}{\delta} \sqrt{\frac{B(T)T}{c_*}}. \quad (6.1)$$

Proof. For any finite collection of qualifying completed transitions, their distances sum to at least their number times δ . Their costs sum to at most $B(T)$, and their disjoint durations sum to at most T . The finite-prefix argument in Theorem 3 therefore gives

$$K\delta \leq \sqrt{B(T)T/c_*}.$$

The same bound for every finite collection excludes an infinite qualifying collection. $\square$

For an organizational conclusion, introduce an operational distance d_{op} and assume a uniform comparison

$$d_{\text{phys}}(x, y) \geq \alpha_* d_{\text{op}}(x, y), \quad \alpha_* > 0. \quad (6.2)$$

This is an additional physical assumption.

Corollary 2 (Fixed operational resolution). *Under the assumptions of Corollary 1 and (6.2),*

$$K_\delta^{\text{op}}(T) \leq \frac{1}{\alpha_* \delta} \sqrt{\frac{B(T)T}{c_*}}, \quad (6.3)$$

where $K_\delta^{\text{op}}(T)$ counts completed transitions with $d_{\text{op}}(x_n, x_{n+1}) \geq \delta$.

Proof. Every counted transition has $d_{\text{phys},n} \geq \alpha_* \delta$. Apply Corollary 1. $\square$

Thus, at every fixed operational resolution, finite-time infinite organizational change is excluded under the stated energy, kinetic, and metric-comparison assumptions.

The metric comparison cannot be omitted from this organizational interpretation. A speed limit in a transport or thermodynamic metric does not automatically constrain an abstract organizational label.

The counting argument applies to any disjoint sequential partition for which the cost budget and speed–distance law hold. In the uniform-coefficient case, grouping adjacent transitions preserves the corresponding bound by the triangle inequality and Cauchy–Schwarz. Arbitrary subdivision therefore does not create an unlimited number of fixed-resolution changes.

7 Variable kinetic coefficients and the remaining boundary

The uniform lower bound on c_n is not needed for a weighted result.

Corollary 3 (Weighted finite-action bound). *If (5.1) holds with $c_n \geq 0$ and $\mathcal{E}, \mathcal{T} < \infty$, then*

$$\sum_n \sqrt{c_n} d_{\text{phys},n} \leq \sqrt{\mathcal{E}\mathcal{T}} < \infty. \quad (7.1)$$

Proof. For every finite k ,

$$\sum_{n < k} \sqrt{c_n} d_{\text{phys},n} \leq \sum_{n < k} \sqrt{\varepsilon_n \tau_n} \leq \sqrt{\mathcal{E}\mathcal{T}}.$$

Take the limit. □

If $d_{\text{phys},n} \geq \delta > 0$ for all counted transitions, then

$$\sum_n \sqrt{c_n} \leq \frac{1}{\delta} \sqrt{\mathcal{E}\mathcal{T}} < \infty, \tag{7.2}$$

and therefore $c_n \rightarrow 0$.

Conversely, if $c_n \geq c_* > 0$, then $\sum_n d_{\text{phys},n} < \infty$ and hence $d_{\text{phys},n} \rightarrow 0$.

These two implications do not establish the unrestricted disjunction “ $c_n \rightarrow 0$ or $d_{\text{phys},n} \rightarrow 0$.” The factors can alternate. For normalized scalar sequences, take

$$(c_n, d_n) = \begin{cases} (1, 2^{-n}), & n \text{ even,} \\ (4^{-n}, 1), & n \text{ odd.} \end{cases}$$

Then $\sqrt{c_n} d_n = 2^{-n}$ is summable, while neither factor converges to zero. This is an algebraic counterexample to an overstrong limiting statement, not a physical escape mechanism.

The exact general condition is (7.1). In particular, for any fixed positive c_0, d_0 , only finitely many transitions can simultaneously satisfy $c_n \geq c_0$ and $d_{\text{phys},n} \geq d_0$.

The interpretation of coefficient collapse is model-dependent. It can correspond, in an appropriate formulation, to increasing mobility or dynamical activity, decreasing friction, changing control constraints, or the failure of the assumed speed-limit description. These interpretations are not interchangeable.

Nor is an arbitrarily weakened bound evidence of physical coefficient collapse. If a model admits a valid uniform positive coefficient, replacing it with smaller numbers does not remove the no-go result. A proposed escape must explain why no applicable uniform control survives under a fixed physical metric and consistent accounting.

8 Sharpness example: finite-time infinite refinement of a finite trajectory

Example 1 (Bounded-power refinement). Consider the deterministic overdamped model

$$\zeta \dot{x} = F, \quad \zeta > 0, \quad F > 0,$$

on an interval containing $[0, L]$. Its speed is $v = F/\zeta$ and its dissipative power is

$$P = Fv = \zeta v^2.$$

Define refinement labels

$$x_n = L(1 - 2^{-n}).$$

The dynamics contains no level-dependent acceleration coefficient. The physical increments, crossing durations, and dissipated costs are

$$d_n = L2^{-(n+1)}, \quad \tau_n = \frac{d_n}{v}, \quad \varepsilon_n = Fd_n.$$

Consequently,

$$\sum_n \tau_n = \frac{L}{v}, \quad \sum_n \varepsilon_n = FL, \quad \sum_n d_n = L,$$

and

$$\varepsilon_n \tau_n = \zeta d_n^2.$$

The finite-action length bound is saturated.

The example produces infinitely many labelled transitions that are successively faster and cheaper at bounded power. It therefore shows why finite time and power cannot exclude infinite indexing.

Nevertheless,

$$x_n \rightarrow L,$$

and the physical trajectory extends regularly through $x = L$. No state variable or physical velocity diverges.

The choice of refinement labels is deliberate: an ordinary finite transformation has been partitioned into infinitely many increasingly small changes. Treating each label as a separately recorded event would add a controller or record-keeping problem not contained in the example.

This is a continuum sharpness example, not a construction of infinitely many reliable organizational innovations. It shows that the fixed-resolution condition is essential, while making no claim that arbitrarily fine labels remain distinguishable under bounded measurement resources.

9 Operational distinguishability and measurement resolution

An operational distinction must specify the allowed observation. For a concrete classical readout, let retained states x, y produce probability distributions p_x, p_y . With equal priors, the optimal discrimination error is

$$P_{\text{err}} = \frac{1 - d_{\text{TV}}(p_x, p_y)}{2}, \quad d_{\text{TV}}(p, q) = \frac{1}{2} \|p - q\|_1. \quad (9.1)$$

A requirement $P_{\text{err}} \leq \epsilon < 1/2$ therefore implies

$$d_{\text{TV}}(p_x, p_y) \geq 1 - 2\epsilon > 0.$$

More generally, fix a class $\mathcal{M}$ of admissible measurement protocols and define

$$d_{\text{op}}(x, y) = \sup_{M \in \mathcal{M}} d_{\text{TV}}(p_x^M, p_y^M).$$

This may be a pseudometric: physically different states can be indistinguishable to every permitted measurement. It becomes a metric after identifying operationally indistinguishable states.

The class $\mathcal{M}$ must specify relevant bounds on observation time, apparatus, precision, repetitions, and error tolerance. A distinction that becomes accessible only by letting these resources diverge is not a distinction at fixed operational resolution.

Definition 1 (Resolution-robust organizational escape). A retained-transition sequence exhibits resolution-robust organizational escape if there exist $\delta > 0$ and $T_* < \infty$ such that infinitely many completed retained transitions before T_* satisfy

$$d_{\text{op}}(x_n, x_{n+1}) \geq \delta$$

under one fixed admissible measurement standard.

Definition 2 (Vanishing-resolution infinite depth). An infinitely indexed retained-transition sequence has vanishing-resolution depth if

$$d_{\text{op}}(x_n, x_{n+1}) \rightarrow 0.$$

Vanishing-resolution depth may describe refinement toward a finite state, as in Example 1. This is a possibility, not a universal conclusion about all infinite-dimensional systems.

Operational distinguishability does not require every admissible test to be performed. If measurement, verification, or record updates are actually used to generate or retain the sequence, however, their costs and timing belong inside the physical model.

10 Physical versus organizational metrics

The physical metric in a speed limit and the operational metric of organizational distinction need not coincide.

A uniform comparison $d_{\text{phys}} \geq \alpha_* d_{\text{op}}$ transfers the physical no-go result immediately. One sufficient mechanism is bounded sensitivity of readout to physical changes.

For example, suppose physical states are probability measures μ, ν on a spatial metric space. For every admissible measurement M and output event A , suppose the conditional readout probability

$$g_{M,A}(x) = \Pr_M(A \mid x)$$

has Lipschitz constant at most L_{readout} , uniformly in M, A . A coupling argument gives

$$d_{\text{op}}(\mu, \nu) \leq L_{\text{readout}} W_1(\mu, \nu).$$

For measures with finite second moments, $W_1 \leq W_2$, so a speed limit in W_2 supplies

$$d_{\text{phys}} = W_2 \geq \frac{1}{L_{\text{readout}}} d_{\text{op}}.$$

Uniform readout sensitivity is an explicit assumption here, not a consequence of the definition of organization.

Metric decoupling

On transitions with $d_{\text{op},n} > 0$, define

$$\alpha_n = \frac{d_{\text{phys},n}}{d_{\text{op},n}}.$$

If operationally significant differences remain bounded below while $\alpha_n \rightarrow 0$, their physical separation collapses relative to their operational separation. We call this *metric decoupling*.

For a sequence with $d_{\text{op},n} \geq \delta > 0$, Corollary 3 implies

$$\boxed{\sum_n \sqrt{c_n} \alpha_n \leq \frac{1}{\delta} \sqrt{\mathcal{ET}} < \infty.} \tag{10.1}$$

Thus:

- a uniform lower bound on α_n forces $\sum_n \sqrt{c_n} < \infty$;
- a uniform lower bound on c_n forces $\sum_n \alpha_n < \infty$;
- without either lower bound, only the joint weighted condition follows.

Metric decoupling is a candidate way to evade the sufficient no-go assumptions, not an established mechanism. It may reflect genuine causal leverage, omitted downstream machinery, increasing measurement resources, an expanding state description, or a metric that loses sensitivity to the relevant physical differences.

Changing coordinates or using level-dependent normalizations cannot establish an escape. Metrics, units, and coefficients must be transformed consistently. The meaningful object is the physical inequality, including its weighted distance, rather than one coefficient viewed in isolation.

Weak convergence and stronger observables

Finite accumulated distance makes (x_n) a Cauchy sequence. If the relevant physical metric space is complete, it converges to a limit in that space. This concerns the retained-state sequence; controlling every intermediate excursion requires an appropriate trajectory-length or regularity bound.

Moreover, convergence in a weak metric does not imply boundedness of stronger observables. Let

$$x^{(N)} = (2^{-1}, 2^{-2}, \dots, 2^{-N}, 0, 0, \dots).$$

Then

$$\|x^{(N+1)} - x^{(N)}\|_{\ell^2} = 2^{-(N+1)},$$

so the accumulated ℓ^2 distance is finite and $x^{(N)}$ converges in ℓ^2 . But

$$Q(x^{(N)}) = \sum_{j \geq 1} 4^j |x_j^{(N)}|^2 = N \rightarrow \infty.$$

The limit belongs to ℓ^2 but not to the space where this weighted quadratic observable is finite.

Such distinctions matter for PDEs and cascades [14, 13]. A limit may be regular in an ambient weak space and singular in a physically relevant stronger class.

The appropriate question is therefore whether the stronger organizational observable is operationally meaningful and resolvable with bounded physical resources. Finite metric length alone does not answer it.

11 Landauer, reversible computation, and other special cases

The finite-action theorem does not assume that every organizational transformation erases information. Landauer's principle supplies one conditional energetic floor; it is not the foundation of the general no-go result.

Landauer as a conditional floor

Under an appropriate isothermal erasure protocol, a net reduction of memory entropy by I_n bits gives a work or heat bound of the form

$$\langle W_n \rangle \geq k_B T_n \ln 2 I_n.$$

Its interpretation requires the memory Hamiltonian, initial correlations, accessible side information, bath, and auxiliary records to be accounted for [3, 6].

For example, when endpoint internal energies agree and the controller is restored, the work and bath-heat bounds coincide. If additionally

$$T_n \geq T_{\min} > 0, \quad I_n \geq I_{\min} > 0,$$

a positive mean erasure-work floor follows.

Organizational change does not automatically satisfy the erasure premise. Reversible computation may preserve information in intermediate records and later remove those records through uncomputation [4, 5].

Landauer work or heat must also be distinguished from excess dissipation. For ideal isothermal erasure,

$$\Delta F_n = k_B T_n \ln 2 I_n,$$

while

$$W_{\text{diss},n} = \langle W_n \rangle - \Delta F_n$$

can approach zero in the reversible limit.

Finally, an average erasure bound is not a fixed pathwise charge. To use it in a duration argument, one may consider prescribed protocols with deterministic durations and bounded gross mean power supplying their required work. To use it in a cumulative pathwise budget, an appropriate pathwise charged-cost premise must be justified separately.

Classical stochastic speed limits

A concrete model-specific speed–distance inequality can be derived for Markov jump dynamics. For forward and reverse probability fluxes a_{ij}, b_{ij} on undirected edges, define

$$A(t) = \sum_{i < j} (a_{ij} + b_{ij})$$

and dimensionless entropy-production rate

$$\sigma(t) = \sum_{i < j} (a_{ij} - b_{ij}) \ln \frac{a_{ij}}{b_{ij}}.$$

For positive fluxes, the elementary inequality

$$(a - b) \ln(a/b) \geq \frac{2(a - b)^2}{a + b}$$

gives

$$\|\dot{p}(t)\|_1 \leq \sqrt{2A(t)\sigma(t)}.$$

Integration and Cauchy–Schwarz yield

$$d_{\text{TV}}(p_0, p_\tau)^2 \leq \frac{1}{2} \left(\int_0^\tau A(t) dt \right) \left(\int_0^\tau \sigma(t) dt \right).$$

Consequently, if $A(t) \leq A_{\max}$ and $\Sigma = \int_0^\tau \sigma(t) dt$,

$$\Sigma \tau \geq \frac{2}{A_{\max}} d_{\text{TV}}(p_0, p_\tau)^2.$$

With an isothermal thermodynamic interpretation and $\varepsilon_{\text{diss}} = k_B T \Sigma$,

$$\varepsilon_{\text{diss}} \tau \geq \frac{2k_B T}{A_{\max}} d_{\text{TV}}(p_0, p_\tau)^2.$$

This illustrates the stochastic speed-limit structure [7]. The coefficient depends on the activity bound and on the thermodynamic interpretation. Bounded incoming power alone does not bound dynamical activity. The energetic budget must apply to the same dissipative quantity used in the inequality.

Overdamped transport and thermodynamic length

For an appropriate isothermal overdamped diffusion with constant friction $\zeta > 0$, probability density p , and probability-current velocity v , the dissipative action has the form

$$\varepsilon_{\text{diss}} = \zeta \int_0^\tau \int p |v|^2 dx dt.$$

For admissible probability curves with finite second moments, the transport-action inequality gives

$$\varepsilon_{\text{diss}} \tau \geq \zeta W_2(p_0, p_\tau)^2.$$

This connects finite-time thermodynamics with optimal transport [8].

The resulting physical metric is a transport metric. Its relation to operational distinguishability must still be established, as in the preceding section.

Thermodynamic length provides related bounds in suitable slow-driving regimes [9, 10]. Those approximations must not be extrapolated without justification into a proposed limit of vanishing transition times. Metric degeneracy or changing friction can also alter the uniformity assumptions.

Reversible dynamics and quantum speed limits

Logical reversibility does not by itself imply arbitrarily fast computation. Conversely, zero thermodynamic entropy production does not imply zero physical motion.

For example, isolated Hamiltonian evolution need not obey a positive dissipative inequality of the particular form (5.1). Quantum speed limits constrain evolution using quantities such as energy above a ground state or energy uncertainty under their respective assumptions [11].

Such stored dynamical resources are not automatically additive energetic costs consumed anew at every transition. They cannot be substituted for ε_n without an accounting argument. Bounded generator strength or energy uncertainty can supply other metric-speed bounds, but these remain model-specific statements.

The no-go boundary is therefore a family of conditional results, not a claim that one dissipation formula covers all physical implementations.

12 Implications for Organizational Accessibility

Organizational Accessibility separates retained architecture, candidate exposure, the variation process, establishment, and access to organization developed elsewhere [1]. This decomposition remains useful, but its singularity interpretation must respect the bounds established here.

In the sequential stochastic model, $\lambda_n = N_n \nu_n p_n$ reveals precisely why the stated processor and rate bounds close C. Better establishment probability cannot grow without bound, and architectural improvements cannot restore explosion while all hypotheses of Theorem 1 remain valid.

In the continuum cascade, increasing specific turnover may result from shrinking transformed quantities. A proposed organizational interpretation must therefore identify the actual retained states, their transition durations, charged costs, physical distances, and operational distinctions. The stationary ratio e_n/Φ_n supplies none of these identifications automatically.

Reuse, templating, catalytic machinery, recombination, and horizontal transfer can reduce the work that remains to be performed. They do not establish a fixed-resolution escape unless the resulting transformations satisfy the full physical accounting. Access, transport, integration, control, and retention can remain consequential even when reconstruction of prior structure is avoided.

The ordinary cumulative claim is not threatened by this boundary. Retained structure can improve access to later targets without generating infinitely many targets in finite time. The interventional criterion for cumulative contribution in Organizational Accessibility remains a separate question from finite-time organizational escape.

13 Novelty boundary

None of the following is claimed as new generic mathematics: the pure-birth criterion, bounded-rate non-explosion, geometric-series tests for stationary cascades, Cauchy–Schwarz bounds, finite-length Cauchy convergence, thermodynamic speed limits, or Zeno accumulation [2, 7, 8, 12]. Nor is finite material alongside stronger-norm divergence a new singularity mechanism [13].

The proposed contribution is an application and clarification:

1. the stated physical processor assumptions already close the stochastic accessibility route;
2. bounded throughput and bounded specific turnover answer different questions;
3. a physically meaningful organizational escape requires an operational resolution standard;
4. the physical no-go theorem transfers to organization only through an explicit metric comparison;
5. the remaining necessary condition concerns a weighted physical cost scale, not an organizational index alone.

Whether this combination constitutes a distinctive contribution relative to all existing literature requires a separate source audit. No priority claim is made here.

14 Open problem

The remaining question is narrower than whether a prescribed rate sequence can explode.

Can an autonomous organizational system generate successive retained transformations that remain operationally nontrivial, while their physical speed–distance cost scale collapses sufficiently fast for finite-time infinite depth?

A candidate must realize actual dynamics and accounting with

$$d_{\text{op},n} \geq \delta > 0, \quad \sum_n \tau_n < \infty, \quad \sum_n \varepsilon_n < \infty,$$

while necessarily satisfying

$$\sum_n \sqrt{c_n} d_{\text{phys},n} < \infty. \quad (14.1)$$

It must do so without concealing an unbounded resource in mobility, activity, precision, control bandwidth, memory, measurement, or an omitted subsystem.

Equation (14.1) is a necessary feasibility condition, not an equivalent existence theorem. It does not supply the dynamics, retention, or budget required by a constructive answer.

Under uniform kinetic control, the candidate must demonstrate the required physical–operational metric decoupling. Under uniform metric comparison, it must demonstrate the required kinetic-coefficient collapse. When both vary, the joint weighted condition is decisive.

The present paper does not conjecture that such a system exists. A successful construction would have to identify its collapsing scale endogenously. An additional no-go theorem could instead derive a uniform bound from a more explicit class of physical implementations.

15 Conclusion

Infinite labelled depth does not establish infinite physical change. Infinite physical variation, when present, does not automatically establish infinitely many retained organizational distinctions at a fixed operational resolution.

The stochastic accessibility route is closed under the stated material and per-realization rate bounds. The continuum cascade survives bounded aggregate throughput through a shrinking inventory and imposed increasing specific turnover, without thereby establishing a physical transient escape.

The central conditional result is:

finite time and finite charged energy + uniform speed–distance control + uniform physical–operational metric comparison $\implies$ finitely many fixed-resolution organizational transitions.

The bounded-power refinement example shows why the resolution condition is essential. The weak-versus-strong norm example shows why finite physical metric length must not be mistaken for a universal regularity theorem.

A physically meaningful counterexample must therefore do more than assign an increasing level number or prescribe increasing rates. It must explain how retained organization maintains operationally significant distinctions while the relevant weighted physical cost scale collapses under complete resource accounting.

Reference status. The supplied Organizational Accessibility v8 manuscript is the source context for this paper. The external references below are a provisional primary-source bibliography carried forward from the existing reference inventory. They have not been freshly retrieved or independently audited for this revision. Their exact bibliographic details and the applicability of model-specific results require the separate literature audit.

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